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Logísticos Binários:

$$\hat{P}_{y=0} = 1 - \hat{P}_{y=1} = 1 - \frac{e^z}{1 + e^z} = \frac{1 - e^z}{1 + e^z}$$

NÃO Evento
ref.

$$\hat{P}_{y=1} = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}$$

Evento
Alt. 1

logito

Soma = 1

$$\hat{z}_i = \alpha + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_K X_{iK}$$

Bernoulli
LL (log Lik)

(MAX Likelihood) MÁXIMA VEROSSIMILHANÇA

Se $\hat{p}_{y=1} \geq \text{cutoff} \rightarrow \text{EVENTO}$ Se $\hat{p}_{y=1} < \text{cutoff} \rightarrow \text{NÃO EVENTO}$

PARA DETERMINADO CUTOFF!

Matriz de Confusão:

		Real	
		True	Falsa
Predicted	True	TP	FP
	Falsa	FN	TN

Prec

Sensit.

Especif.

$$\text{ACURÁCIA} = \frac{TP + TN}{TP + FP + FN + TN}$$

(EGM)

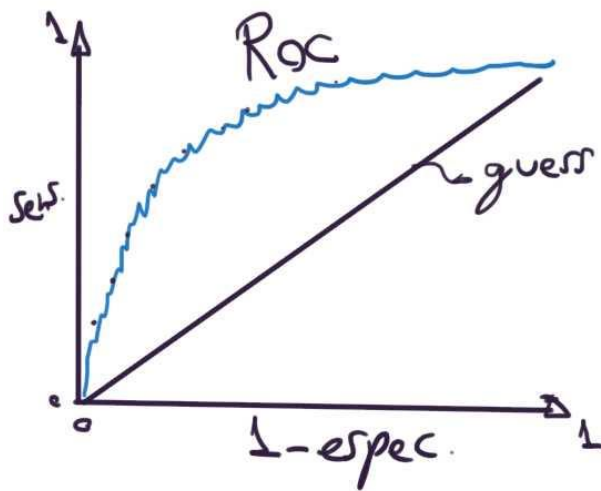
$$\text{Sensitividade} = \frac{TP}{TP + FN}$$

(Recall)

$$\text{Especificidade} = \frac{TN}{TN + FP}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$F_1\text{-Score} = \frac{2 \times \text{Recall} \times \text{Precision}}{(\text{Precision} + \text{Recall})}$$

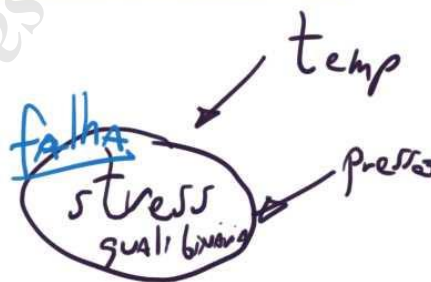


INDEPENDÊ
DO
CUTOFF!

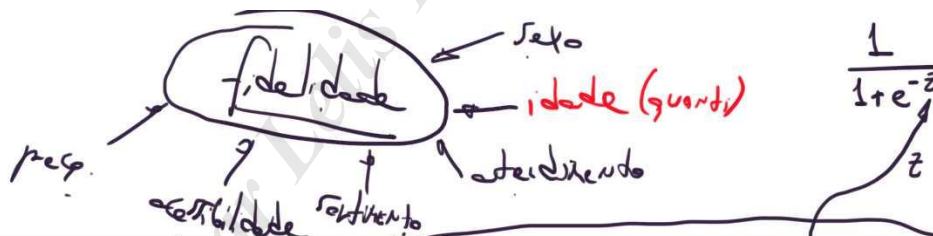
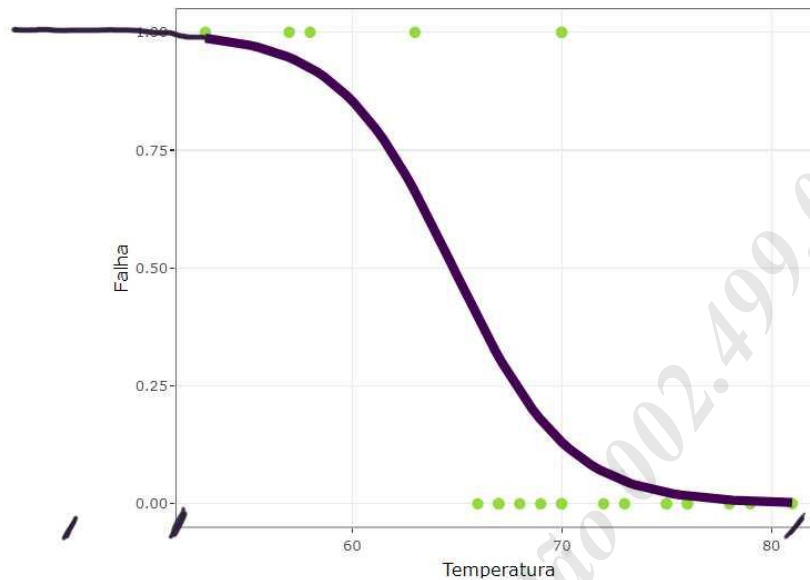
- | | |
|----------------------------------|-------|
| - AUROC | - AIC |
| - GINI = $\frac{Roc - 0,5}{0,5}$ | - BIC |
| - logLik | |

Robert Clemen (1997)

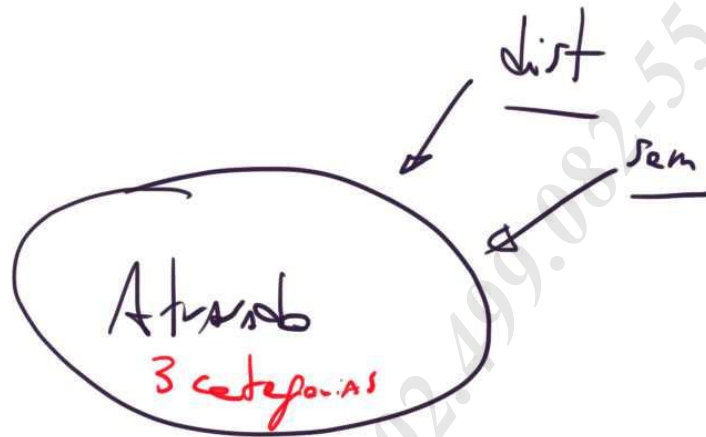
MAKING HARD DECISIONS



$$P_{falha} = \frac{1}{1 + e^{-(23,77 - 0,366 \times temp)}}$$



$$\log \left[\frac{\hat{P}}{1 - \hat{P}} \right] = -69.03 + 1.77(\text{sexo}_{\text{masculino}}) + 1.69(\text{idade}) + 1.68(\text{atendimento}_2) + 1.82(\text{atendimento}_3) + 3.32(\text{atendimento}_4) + 4.33(\text{atendimento}_5) + 1.86(\text{sortimento}_2) + 2.06(\text{sortimento}_3) + 3.34(\text{sortimento}_4) + 5.95(\text{sortimento}_5) + 2.35(\text{acessibilidade}_2) + 2.92(\text{acessibilidade}_3) + 4.85(\text{acessibilidade}_4) + 5.36(\text{acessibilidade}_5) + 2.92(\text{preço}_3) + 3.04(\text{preço}_4) + 3.91(\text{preço}_5)$$

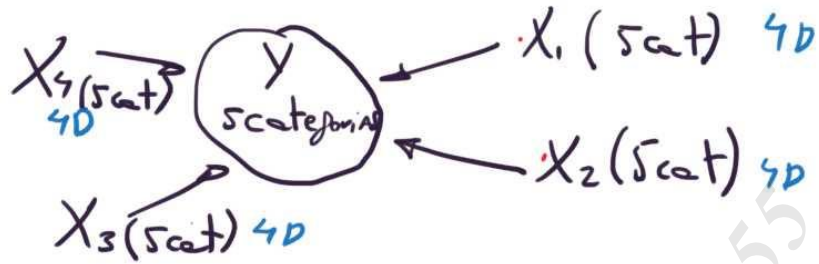


Logística Multinomial

$$Z_1 = \alpha_1 + \beta_{11} \cdot X_1 + \beta_{21} \cdot X_2 + \dots + \beta_{K1} \cdot X_K$$

$$Z_2 = \alpha_2 + \beta_{12} \cdot X_1 + \beta_{22} \cdot X_2 + \dots + \beta_{K2} \cdot X_K$$

$$\begin{aligned} \text{ref } P_0 &= \frac{e^0}{e^0 + e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{z_1} + e^{z_2}} \\ \text{alt1 } P_1 &= \frac{e^{z_1}}{1 + e^{z_1} + e^{z_2}} \\ \text{alt2 } P_2 &= \frac{e^{z_2}}{1 + e^{z_1} + e^{z_2}} \end{aligned} \quad \left. \vphantom{\begin{aligned} P_0 \\ P_1 \\ P_2 \end{aligned}} \right\} \text{SOMA} = 1$$



OLS (ERRADO!!!) } 1 α e 4 β 's (ERRADO!)
SEM DUMMIES.

Log. Multinomial com Dummies (Correto!!!)

4 logitos com 1 α e 16 β 's.

4 α 's e 64 β 's (Correto!)

	Z_1	Z_2
dist	0.557** (0.243)	1.077*** (0.302)
sem	1.667*** (0.575)	2.892*** (0.685)
Constant	-33.069*** (12.147)	-62.216*** (14.643)

$$P_0 = \frac{1}{1 + e^{(-33.07 + 0.55 \cdot \text{dist} + 1.66 \cdot \text{sem})} + e^{(-62.22 + 1.07 \cdot \text{dist} + 2.89 \cdot \text{sem})}}$$

* Z_1 Z_2

$$P_1 = \frac{e^{(-33.07 + 0.55 \cdot \text{dist} + 1.66 \cdot \text{sem})}}{1 + e^{(-33.07 + 0.55 \cdot \text{dist} + 1.66 \cdot \text{sem})} + e^{(-62.22 + 1.07 \cdot \text{dist} + 2.89 \cdot \text{sem})}}$$

MEISMO DENOMINADOR

$$P_2 = \frac{e^{(-62.22 + 1.07 \cdot \text{dist} + 2.89 \cdot \text{sem})}}{1 + e^{(-33.07 + 0.55 \cdot \text{dist} + 1.66 \cdot \text{sem})} + e^{(-62.22 + 1.07 \cdot \text{dist} + 2.89 \cdot \text{sem})}}$$

MEISMO DENOMINADOR

